

CLC - Verification Test Plan - Final

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ESG-452: Capstone Group

Verification Test must include the following

1. Test description

1. Make sure to include which function requirements/design specifications are being tested and any relationship to engineering standards.
2. Materials list
3. Equipment list
 1. Complete with all settings needed to conduct the test
4. Procedures
5. Approval of Mentor and/or faculty expert and instructor

Once you have a complete and approved documentation of the test, work with the lab staff

Michael Dennison and/or the approving expert to schedule time for the test to be conducted.

Center of Mass Test

1. Test description

The purpose of this test is to identify the center of mass of the prototype in X Y and Z coordinate planes. Having an accurate center of mass when compared to an actual medium sized canine was one of the functional requirements of the prototype. Ideally the center of mass should be in the very middle of the upper chest cavity of the dummy and this test will help the team identify when the center of mass sits currently and it will help the team dictate if and where any extra weight will need to be added to the skeletal structure. It will also help the team meet a total weight of around 25 pounds for the dummy which was another one of the prototype's functional requirements.

2. Materials list

- Retort Stand
- Plump line (mass + string)
- Walking Harness
- Rope
- Scissor
- Tape
- Marker
- Leash (with clip at the end)

3. Equipment list

- weight scale

4. Procedures

1. Use the scissor to cut a 20 in string

2. Attach the mass at one end of the string to make a plumb line
3. Mark and label two random points of suspensions on the dummy
4. Attach the leash onto the dummy's harness at suspension 1
5. Hang the dummy up at the first point of suspension on the retort stand
6. Wait until the dummy stop moving/rotating
7. Hang the plumb line in front on the dummy
8. Use a marker to mark 2 points on the dummy along the plumb line
9. Use a small tape to make a line or use a ruler and a marker to draw a line on the dummy jointing two points together
10. Repeat the steps 5 to 9 for all three dimensional planes of the dummy to obtain the center of mass in three directions: x, y and z
11. Hang the dummy up at the second point of suspension
12. Repeat steps 6 to 10
13. Mark the intersection of the two taped/drawn lines => Center of gravity

Drop Test

1. Test description: This test is created to examine the structural integrity of the canine crash test dummy model. This drop test involves the canine crash test dummy being dropped from a predefined height and stopping before it hits a surface. This test will help the team determine if the model can withstand a 30 mile per hour crash. The drop test will also test the toughness of the model.
2. Materials list: The drop test requires materials, such as a rope that is able to withstand more than 100 lbs. force and properly secure the model, a net and a mattress to ensure the safety of the canine crash test dummy in case something does not go according to plan, yellow tape to prevent people from walking into the area where the drop test is taking place, the canine crash test dummy, and the harness to ensure the canine model is secured.
3. Equipment list
 - a. The test will take place at a building that has an area to drop the model at 30 ft.
The area must have enough space for the model to stop and it must not be transited by people.
4. Procedures
 - The test will be performed on the middle section of the Engineering building where there is not stairs but elevator
 - The dummy should be dropped for 30 ft. to achieve 30 mph. and the dummy should be stopped by the harness 5 ft above the ground
 - Estimate the area and the floors that the dummy will be dropped from
 - Set up caution tape around the drop test area to prevent others from getting too close

- Use military knots to tie the rope at 2 ends of the dummy's harness
- The other ends of the ropes will be tied to the beams on the balcony
- On the ground, set up a large mattress pad just in case the canine crash test dummy
- Two members will be in charge of dropping the dummy
- The other two will be on the ground floor to record videos and make sure passengers will not stay far from the testing area

Configuration Test:

1. Test description

The main goal of the project is that the Canine Crash Test Dummy resembles a real dog as much as possible when it comes to certain factors such as the weight, dimensions, and range of motion. The team has selected a Beagle as the basis of the design because of the vast information and data available on the breed, as well as the breed resembling a medium-small size dog. The model must have the correct weight, height, length, and width of that particular breed to meet the required standards. The range of motion of the model must also be comparable to a real canine. As a final doublecheck, the dummy will be tested to make sure it is able to sit and lay down in the positions expected that a dog would in a vehicle harness. The Configuration Test will be essential in determining if the dummy meets these requirements.

2. Materials list

- Weight scale
- A flat and wide piece of wood or metal
- Tape measure
- Marker
- Paper, pen, laptop

3. Equipment list

- Canine Crash Test Dummy
- Weight Scale
- Flat piece of wood
- Tape Measure
- Angle ruler/protractor

4. Procedures

In order to measure the weight of the Canine Crash Test Dummy, a flat piece of wood will be placed on a scale, then the scale will be zeroed. The wood allows the dummy's total weight to be measured because the dummy will have the required surface area to sit easily on the wood instead of the small scale. The dummy's weight will be shown once the value of the scale has stabilized.

For the dimensions, a tape measure will be used. The height will be measured from the floor to the top of the dog's head while in a standing position. The length will be measured from the tip of the dog's nose to the back of the pelvis. The width applies mainly to the chest girth however. Since the chest was bent in a round shape, the width will be the diameter, then the equation of $2 \cdot \pi \cdot r$ will be used to determine if the girth of the chest is within the range of medium sized dogs that the team has researched previously.

The range of motion of the dog will be measured using an angular ruler or a protractor. The various limbs will be placed in the required positions to sit or lay down, then the angles of the joints will be measured to see how they compare to a real canine. The ball and socket joints will also be checked for their rotational abilities in comparison to real ball and socket joints of canines. To double check the configuration, the dummy will be placed in laying and sitting positions under the proper angles and made sure that it is not easily moved out of position under subtle forces.